Lecture 7 2025-09-18

Bisection method

- Assume f is continuous on $[a,b]$ and $f(a)$ and $f(b)$ have opposite signs.
- IVT tells us that under these conditions there exists $p \in (a,b)$ such that $f(p)=0$.

Input: f continuous on interval $[a,b]$ and $f(a)$ & $f(b)$ have opposite signs.

- We set $a_1=a$, $b_1=b$ and $p_1 = \frac{a_1+b_1}{2}$ the midpoint of $[a,b]=[a_1,b_1]$.

- If $f(p_1)=0$, we are done.
- If $f(p_1)\neq 0$ then it has the same sign as either $f(a_1)$ or $f(b_1)$

  * If $f(p_1)$ and $f(a_1)$ have the same sign, then by IVT $p \in (p_1, b_1)$
  Set $a_2=p_1$ and $b_2=b_1$.

  * If $f(p_1)$ and $f(b_1)$ have the same sign, then by IVT there is a root $p \in (a_1, p_1)$.
  Set $a_2=a_1$ and $b_2=p_1$.

Then reapply the procedure to $[a_2,b_2]$.

$(a,b)=(a_1,b_1)$ 1st step

Stopping criteria

(a_2, b_2) 2nd step
(a_3, b_3) 3rd step
⋮
(a_n, b_n) nth step

We used $|p_n - a_n| < tol$

Could also use:
- $|p_n - p_{n-1}| < \varepsilon$
- $\frac{|p_n - p_{n-1}|}{|p_n|} < \varepsilon$
- $|f(p_n)| < \varepsilon$
- or a combination of these

Denote (a_n, b_n) the interval in the nth step

$$p_n = \frac{a_n + b_n}{2}$$

Notice that the width of the intervals is cut in half in each iteration.

Theorem: If $f \in C([a,b])$ and $f(a) * f(b) < 0$. Then the bisection method generates a sequence $\{p_n\}_{n=1}^{\infty}$ approximating a zero p of f with

$$|p_n - p| \leq \frac{b-a}{2^n} \quad \text{when } n \geq 1.$$

Proof:

interval has half the size

$$b_n - a_n = \frac{1}{2}(b_{n-1} - a_{n-1}) = \frac{1}{4}(b_{n-2} - a_{n-2}) = \dots = \frac{1}{2^{n-1}}(b_1 - a_1)$$

$$= \frac{1}{2^{n-1}}(b-a)$$

$$|p - p_n| \leq \frac{1}{2}(b_n - a_n) = \frac{1}{2^n}(b-a)$$

$p \in (a_n, b_n)$ midpoint of (a_n, b_n)

Rate of convergence $p_n = p + \mathcal{O}\left(\frac{1}{2^n}\right)$

↳ exponential convergence rate

How many iterations are needed to achieve $|p_n - p| < tol$?

$$\frac{b-a}{2^n} \leq tol \quad \text{is sufficient}$$

Solve for n:

$$b-a \leq tol \cdot 2^n$$

$$\Longleftrightarrow \quad \frac{b-a}{tol} \leq 2^n$$

$$\Longleftrightarrow \quad \log\left(\frac{b-a}{tol}\right) \leq n \log 2$$

$$\Longleftrightarrow \quad \frac{\log\left(\frac{b-a}{tol}\right)}{\log 2} = \frac{\log(b-a) - \log(tol)}{\log 2} \leq n \quad \text{is sufficient}$$

Example: Compute $\sqrt{2}$ with accuracy of 0.1 starting with $[1,2]$

Solution: $f(x) = x^2 - 2$, $f(1) = -1$, $f(2) = 2$

$\Rightarrow$ f has a root in $[1,2]$, conditions for using bisection are satisfied

iterations needed: $\frac{2-1}{2^n} \leq 0.1$

$\Longleftrightarrow 10 \leq 2^n \Rightarrow n = 4$ sufficient

What do iterates look like?

1.) $p_1 = 1.5 \left(= \frac{2+1}{2} = \frac{a+b}{2}\right)$

$f(1.5) > 0 \quad \Rightarrow \quad b_2 = p_1 = 1.5$

$a_2 = a_1 = 1$

2.) $p_2 = \frac{1.5+1}{2} = 1.25$

$f(1.25) < 0 \quad \Rightarrow \quad a_3 = p_2 = 1.25$

$b_3 = b_2 = 1.5$

3.) $p_3 = \frac{1.25+1.5}{2} = 1.375$

$f(1.375) < 0$

$a_4 = p_3 = 1.375$

$b_4 = b_3 = 1.5$

4.) $p_4 = \frac{1.375+1.5}{2} = \underline{1.4375}$

We used

$$|p_n - a_n| < \varepsilon \quad \text{as a stopping criterion}$$

$$|p_n - a_n| \leq \underbrace{|p_n - p|}_{\leq \frac{(b-a)}{2^n}} + \underbrace{|p - a_n|}_{\leq \frac{(b-a)}{2^{n-1}}} \leq (b-a)\left(\frac{1}{2^n} + \frac{1}{2^{n-1}}\right) = \frac{3(b-a)}{2^{n-1}} = \frac{6(b-a)}{2^n}$$

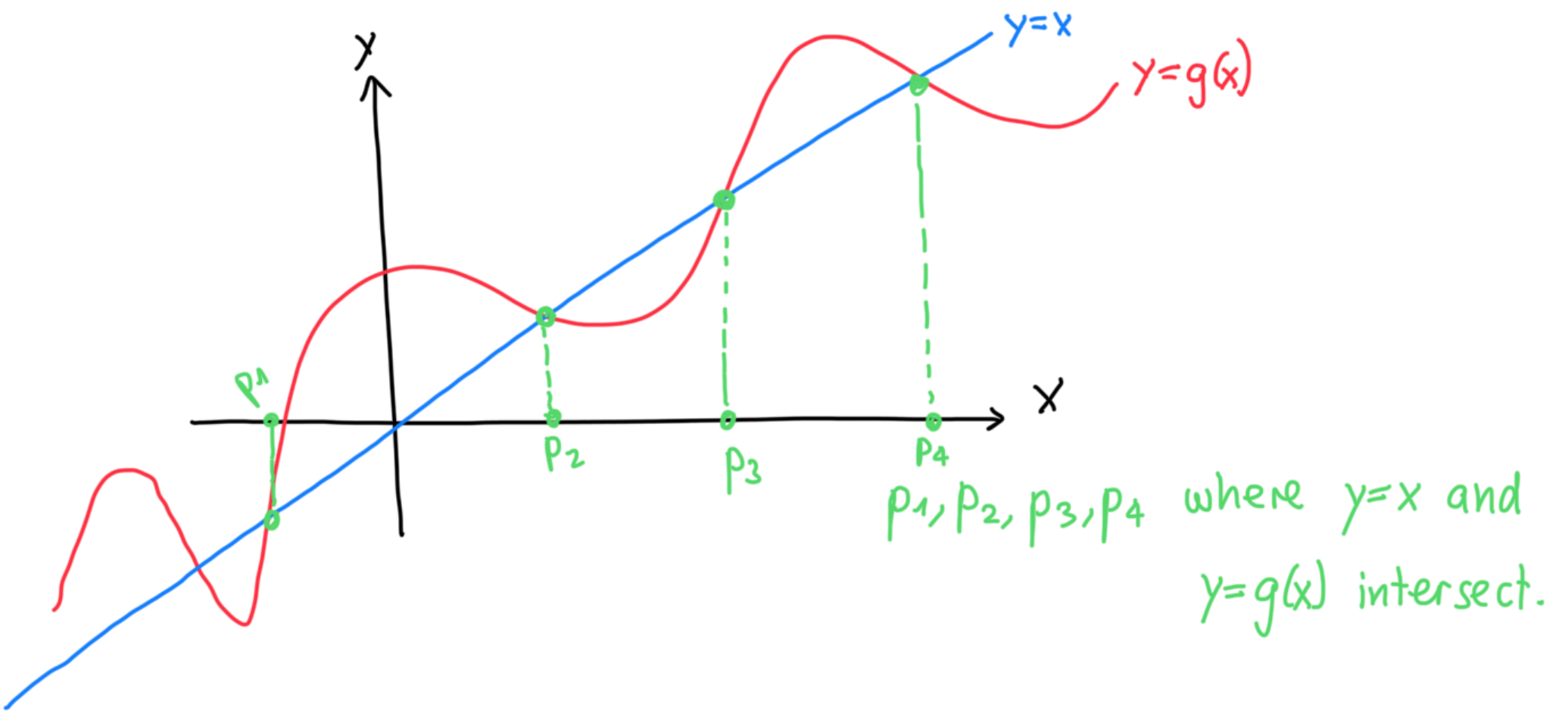
2.2. Fixed point iteration

Def: p is a fixed point of g(x) if g(p)=p.

Idea: To solve $f(x)=0$, define a new function $g(x)$ such that

$g(x)=x$ iff $f(x)=0$

Example: $g(x)=x-f(x)$ equal to x iff $f(x)=0$



p_1, p_2, p_3, p_4 where $y=x$ and $y=g(x)$ intersect.

Sufficient conditions for existence and uniqueness

Sufficient conditions for existence:

1.) $g \in C([a,b])$

2.) g maps [a,b] to [a,b] ($a \leq g(x) \leq b$ for $x \in [a,b]$)